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Predicting patient outcomes and risk for revision surgery after hip and knee replacement surgery: study protocol for a comparison of modelling approaches using the Swiss National Joint Registry (SIRIS)

Léonie Hofstetter[1,2,3], Nathalie Schweyckart[1], Christof Seiler[4,5,6], Christian Brand[7], Laura C. Rosella[8], Mazda Farshad[3], Milo A. Puhan[2] and Cesar A. Hincapié[1,2,3*]

Abstract

Background Prediction of postoperative patient-reported outcomes and risk for revision surgery after total hip arthroplasty (THA) or total knee arthroplasty (TKA) can inform clinical decision-making, health resource allocation, and care planning. Machine learning (ML) algorithms are increasingly used as an alternative to traditional logistic regression (LR) prediction, but there is uncertainty about their superiority in overall model performance. The aim of this study is to compare the predictive performance of LR with different ML approaches for predicting patient outcomes and risk for revision surgery after THA and TKA.

Methods A population-based historical cohort study will be developed using routinely collected data from all primary and revision THA and TKA procedures performed in Switzerland and registered in the Swiss National Joint Registry (SIRIS). Patients of age ≥ 18 years with surgery for primary osteoarthritis from 01 January 2015 up to 31 December 2023 will be included. Outcomes of interest will be (1) 12-month postoperative poor pain outcome (defined as < 50% improvement of pain or < 3 absolute reduction in pain on a 11-point (0 to 10) numeric rating scale) and poor satisfaction outcome, and (2) early revision within 5 years after primary surgery. Prespecified predictor variables will include demographic characteristics, comorbidity score, patient-reported health status measures, and surgical variables. Measures of overall predictive accuracy, discrimination, and calibration will be used to compare predictive performance, and decision curve analysis performed to evaluate the clinical usefulness of models. The models will be internally validated using cross-validation and externally validated using geographical validation. Development of the models will be informed by the updated Transparent Reporting of a multivariable prediction model for Individual Prognosis or Diagnosis (TRIPOD + AI) statement.

Discussion This study will develop, validate, and compare prediction models for postoperative patient-reported outcomes and risk for revision surgery after THA and TKA using SIRIS data.

*Correspondence:
Cesar A. Hincapié
cesar.hincapie@uzh.ch
Full list of author information is available at the end of the article

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Keywords Hip arthroplasty, Knee arthroplasty, Prognostic modelling, Model development, Risk prediction, Machine learning, Decision support techniques

Background

Total hip arthroplasty (THA) and total knee arthroplasty (TKA) are commonly performed surgical procedures in Switzerland [1]. When conservative, non-surgical treatments prove ineffective in managing arthritis symptoms, these procedures provide patients with improved quality of life and function [2]. The Swiss National Joint Registry (SIRIS) was implemented in 2012, and now records about 98% of all hip and knee arthroplasties and accounts for an estimated 22,000 hip joint and 18,000 knee joint replacements each year [3]. While joint replacements typically have an expected lifespan of at least 25 years on average, a small proportion will fail within a shorter time frame and result in poor response outcome or require revision surgery [4, 5]. Despite being highly successful with respect to prosthesis-related outcomes, about 7% to 23% of patients report unfavorable pain outcome after THA, and 10% to 34% after TKA [6]. Premature revision surgery imposes significant challenges for both patients, health systems, and society [7].

Shared decision-making, where clinicians and patients share available evidence and preferences to support upcoming treatment decisions, is known to improve care and costs [8]. Providing patients and clinicians with useful and accurate predictions of postoperative outcomes can facilitate communication and shared decision-making. By capturing patient perspectives on their health status in a standardized way, patient-reported outcome measures (PROMs) are considered important tools to evaluate treatment outcomes [9] and promote healthcare quality improvement [10]. Predicting PROMs and the risk of revision surgery may enhance a patient's awareness of potential risks associated with elective surgery, and also facilitate clinicians' and hospitals' ability to anticipate postoperative courses, healthcare burden, resource allocation, and care planning [11, 12].

There is growing interest in the development and use of clinical prediction models where decisions regarding treatment choices and plans of management are informed by an individual's profile of prognostic factors [13]. While traditional regression-based risk prediction models, for example logistic regression (LR), are widely used, machine learning approaches are increasingly being used as an alternative method [14, 15]. Machine learning models can directly and automatically learn from new data and make decisions or predictions without being explicitly programmed to perform specific tasks [16].

Supervised machine learning models use algorithms that are trained to learn from data in situations when the outcome of interest is specified and then make decisions based on similar characteristics or variables from new data [17]. They are considered powerful tools that may support clinical decision-making by their ability to integrate a large amount of data and to identify complex patterns [18]. Despite the possible advantages of machine learning models, there is uncertainty whether they have superior predictive performance compared to traditional methods such as logistic regression, and more research with better reporting of both types of algorithms are needed to better compare their performance [19]. It has been estimated that only 40% of machine learning models developed for predicting outcomes after THA or TKA have been evaluated in the context of traditional statistical prediction techniques such as logistic regression, and only 30% of studies used external validation [20].

Among machine learning approaches, random forest (RF) and gradient-boosted machines such as extreme gradient boosting (XGBoost)—both regression-based decision trees—are most commonly used in the area of THA and TKA prediction modelling and are well known for their good predictive performance [20]. Despite possible advantages, there is little evidence for their adoption in clinical practice [21]. Some limitations are that concepts such as machine learning are still novel to many health researchers and clinicians, leading to skepticism about their practical application.

A recently updated Cochrane review on the effectiveness of decision support tools showed that they can largely increase patients' knowledge of options, feeling informed and clear about what matters most to them, and their active role in decision-making [22]. With better insights into the risk of experiencing a poor outcome or early revision, patients with an elevated risk for a poor outcome may be identified and benefit from more intensive follow-up and care processes after THA or TKA. Clinical decision-making and health communication might also be improved. To the best of our knowledge, no study has compared RF and XGBoost with LR for prediction modelling of patient-reported outcomes and risk for revision surgery in the orthopaedic literature.

In this study, we aim to compare the predictive performance of LR with different machine learning approaches for predicting patient outcomes and risk for revision surgery after THA and TKA.

Our specific objectives are:

1) To compare the performance of random forest and extreme gradient boosting algorithms with a logistic regression-based model for predicting risk for poor patient-reported outcomes (pain intensity and satisfaction) one year after THA or TKA.
2) To compare the performance of random forest and extreme gradient boosting algorithms with a logistic regression-based model for predicting risk for revision surgery within 5 years after THA or TKA.

Methods

Prediction modelling

Our study will be informed by the prognosis research study (PROGRESS) framework [13], and reported according to the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement [23] and the updated Transparent Reporting of a Multivariable Prediction Model for Individual Prognosis or Diagnosis (TRIPOD + AI) statement, which provides guidance for reporting clinical prediction models that use regression or machine learning methods [24, 25].

Study design and data source

A population-based historical cohort study will be conducted using routinely collected data from all primary and revision hip and knee arthroplasty procedures performed in Switzerland and registered in SIRIS. To ensure the completeness and consistency of SIRIS data, systematic software-generated validation tests are conducted on the received data. Additionally, on a quarterly basis, the registry's statistician employs an automated analysis script to generate master files and lists identifying potential data errors. These findings are then reviewed by the data monitoring team, which conducts root cause analyses for confirmed issues and provides feedback to the respective hospitals.

Eligibility criteria

The study population will consist of patients of age $\geq$ 40 years at the time of surgery who were registered in the SIRIS as having undergone total hip or knee arthroplasty for primary osteoarthritis from 01 January 2015 up to 31 December 2023.

Outcome measures and variables

Objective 1: 12-month postoperative patient-reported outcomes of pain intensity and satisfaction will be the outcome measures for objective 1. Since 2019, all hospitals receiving public funding for elective hip and knee arthroplasties in the Canton of Zurich must collect the Swiss orthopaedic minimal dataset (MDS) [26, 27] consisting of the 5-dimension, 5-level version of the EuroQol health-related quality of life measure (EQ-5D-5L) and a small selection of additional questions on joint-specific pain and patient satisfaction. Questionnaires are collected approximately 1 week preoperatively and 12 months postoperatively. Self-rated health is assessed using the visual analogue scale (VAS) of the EQ-5D-5L ranging from 0 (worst imaginable health) to 100 (best imaginable health). Joint-specific pain intensity (average over last week) is measured on an 11-point numeric rating scale (NRS; 0 being "no pain," and 10 being "the worst pain I can imagine"). Joint-specific satisfaction is assessed on a 5-item scale (1 being "very satisfied," and 5 being "very dissatisfied").

In the absence of consensus for a dichotomous definition of poor response after THA and TKA [28], we were guided by the recommendations of an international Delphi study and defined poor responders for satisfaction outcome as those who answered "very unsatisfied," "unsatisfied," or "neither satisfied nor unsatisfied" [29]. For pain outcome, we defined poor responders as those who reported either a relative change of < 50% improvement in pain intensity from preoperative to 1−year postoperative measurement, or < 3 absolute reduction in pain intensity on the 11-point NRS.

Objective 2: The outcome of interest for objective 2 will be early revision after THA or TKA. This is defined as revision surgery for any indication within five years after primary surgery.

Predictor variables

To prespecify a set of predictor variables, we followed the approach recommended by Steyerberg et al. [30]. First, candidate predictor variables were identified from the available relevant literature. Then, the candidate predictor variables were shared with and discussed in the interdisciplinary research group. Final selection of predictor variables was made through discussion and consensus of all collaborators. The predictor variables that were identified and that will be included in the model are summarised in Tables 1 and 2. For objective 2, PROMs will not be included because they are only collected since 2019, which would lead to a substantial decrease of the sample size. Nominal categorical variables will be transformed to dummy variables using one-hot encoding, and ordinal categorical variables will be transformed to numerical variables.

Development and external validation cohorts

For poor patient-reported outcome prediction modelling, the development and validation cohorts will be formed based on the 12 districts of Canton Zurich. The development cohort will include the districts of Zürich, Affoltern,

Table 1 Summary of selected predictor variables for patient outcomes one year after THA and TKA

Predictor variable	Definition	Variable type	Unit/categories	Evidence
Age	Age at intervention date	Continuous	Years	[31, 32]
Sex	Sex	Binary	1 = female 2 = male	[31–33]
BMI	BMI	Continuous	kg/m^2	
Comorbidities	ASA score	Categorical (ordinal)	1 = no disturbance 2 = mild/moderate disturbance 3 = severe disturbance 4 = life-threatening 5 = moribund	[31, 33, 34]
Preoperative walking ability	Charnley class	Categorical (nominal)	1 = unilaterally diseased 2 = bilaterally diseased 3 = bilaterally diseased, opposite hip or knee prosthesis 4 = other condition affecting walking	[35, 36]
Surgeon volume	Number of surgeries performed by the treating surgeon in the calendar year before the index date	Continuous	Surgeries/year	[37]
Previous surgery of the affected joint	Previous surgery of the affected joint	Binary	0 = no 1 = yes	
Socioeconomic status	Education level	Categorical (ordinal)	1 = low 2 = middle 3 = high	[38]
Preoperative pain intensity	Pain intensity (average over last week)	Continuous	0–10, 10 = worst	[33, 39]
Preoperative joint-specific satisfaction	Preoperative joint-specific satisfaction	Categorical (ordinal)	1 = very satisfied 2 = somewhat satisfied 3 = neither satisfied nor dissatisfied 4 = somewhat dissatisfied 5 = very dissatisfied	
Preoperative general health	Self-perceived general health	Continuous	0–100	

and Dietikon. The external validation cohort will include the districts of Horgen, Meilen, Uster, Hinwil, Dielsdorf, Bülach, Pfäffikon, Winterthur, and Andelfingen.

For revision risk, the development and validation cohorts will be formed based on the seven major regions of Switzerland using Nomenclature of Territorial Units for Statistics (NUTS) [50]. All cantons of NUTS regions 03–07 (Northwestern Switzerland, Zurich, Eastern Switzerland, Central Switzerland and Ticino) and Lichtenstein will form the development cohort, while all cantons of NUTS regions 01 and 02 (Lake Geneva region and Espace Mittelland) will form the external validation cohort.

Sample size

Following Riley et al. [51], we calculated the minimum sample size using the R package *pmsampsize*. As recommended, we set Nagelkerke's $R2 = 0.15$ (Cox-Snell's adjusted $R2 = 0.09$) and the assumed shrinkage factor equal to 0.9.

For the prediction of poor patient-reported outcome after THA, with an observed prevalence of approximately 9.8% for pain and 7.2% for satisfaction in our study population and 20 predictor parameters, 2434 participants are required for poor pain outcome and 2850 for poor satisfaction outcome.

For the prediction of poor patient-reported outcome after TKA, with an observed prevalence of approximately 16.3% for pain and 15.1% for satisfaction and 20 predictor parameters, 1944 participants are required for poor pain outcome and 1992 for poor satisfaction outcome.

For the prediction of revision surgery within 5 years after THA, with an observed prevalence of approximately 2.7% and 23 predictor parameters, 6158 participants are required. For the prediction of revision surgery within 5 years after TKA, with an observed prevalence of approximately 3.9% and 25 predictor parameters, 5232 participants are required.

Our fixed sample sizes exceed these minimum requirements for the development cohorts. Therefore, we expect our study to be adequately powered. For external validation studies of prognostic models, it has been recommended that the validation cohort has a minimum of 100 (or better 200) events [30, 52]. All validation

Table 2 Summary of selected predictor variables for revision surgery within 5 years after THA and TKA

Predictor variable	Definition	Variable type	Unit/categories	Evidence
Age	Age at intervention date	Continuous	Years	[40, 41]
Sex	Sex	Binary	1 = female 2 = male	[40–42]
BMI	BMI	Continuous	kg/m^2	[40, 41]
Comorbidities	ASA score	Categorical (ordinal)	1 = no disturbance 2 = mild/moderate disturbance 3 = severe disturbance 4 = life-threatening 5 = moribund	[41, 43]
Preoperative walking ability	Charnley class	Categorical (nominal)	1 = unilaterally diseased 2 = bilaterally diseased 3 = bilaterally diseased, opposite hip prosthesis 4 = other condition affecting walking	[41]
Surgeon volume	Number of surgeries performed by the treating surgeon in the calendar year before the index date	Continuous	Surgeries/year	[37, 42]
Previous surgery of the affected joint	Previous surgery of the affected joint	Binary	0 = no, 1 = yes	[41]
Surgical variables: THA				
Approach	Surgical approach	Categorical (nominal)	1 = anterior 2 = anterolateral 3 = lateral 4 = posterior 5 = transfemoral 6 = other approach 7 = trochanter osteotomy	[44–46]
Component fixation	Fixation of surgical components	Categorical (nominal)	1 = all cemented 2 = all uncemented 3 = acetabular component (AC) uncemented, femoral component (FC) cemented 4 = AC cemented, FC uncemented 5 = reinforcement ring, FC cemented 6 = reinforcement ring, FC uncemented	[46]
Surgical variables: TKA				
Prosthesis	Type of prosthesis	Categorical (nominal)	1 = bicruciate retaining 2 = posterior cruciate retaining 3 = cruciate sacrificing/ultracongruent (UCOR) 4 = posterior stabilized 5 = medial pivot 6 = semi-constrained/constrained 7 = hinge type 8 = other arthroplasty	[47]
Subtype	Bearing of prosthesis	Binary	1 = mobile bearing 2 = fixed bearing	[47]
Patellar component	Patella resurfacing	Categorical (nominal)	1 = yes 2 = no 3 = prior patellectomy	[48]
Component fixation	Fixation of surgical components	Categorical (nominal)	1 = femur + tibia cemented 2 = femur + tibia uncemented 3 = femur uncemented, tibia cemented 4 = femur cemented, tibia uncemented	[49]

cohorts exceed the recommended number of events for validation.

Missing data

If an outcome measurement is missing, the patient data will be excluded from the analysis. Around 60% of patients had to be excluded because of missing pain or satisfaction outcome. All populations with or without patient-reported outcome were comparable (see Supplementary appendix eTables 1 and 2). Predictors with missingness exceeding 30% will be excluded. Missing data among the remaining predictors will be assumed missing at random. We will perform multiple imputations using the *mice* package in R, which imputes incomplete multivariate data by chained equations algorithm [30, 53]. Five imputed datasets will be generated. The model will be developed in each imputed dataset and averaged over the estimated predictions.

Since for some patients the study ends before potentially undergoing revision surgery, the data will include right-censored observations. To address this, Kaplan–Meier imputation will be performed using the *CondiS* package in R. This method imputes survival times for censored cases based on their conditional survival distributions, derived from the Kaplan–Meier estimator [54].

Model development

We will develop the prediction models based on the "full model approach," which is using the pre-selected prognostic factors without any being excluded at a later stage. To avoid loss of prognostic information, continuous variables will be kept continuous and not categorised [55].

LR model

We will use multivariable logistic regression to develop a prediction model for the incidence of revision surgery within 5 years after primary surgery. For predicting poor patient-reported outcome, outcomes of interest will be dichotomized.

To account for the risk of overfitting, an additional penalized logistic regression model will be developed using the *glmnet* package in R. Elastic Net regression, a combination of Ridge and least absolute shrinkage and selection operator (LASSO) regression, will be used.

RF and XGBoost model

Both RF and XGBoost algorithms are commonly used for prediction modelling in the area of THA and TKA [20]. RF operates by constructing a multitude of decision trees during the training phase, each using a subset of the development data. The individual predictions of all trees are aggregated to produce the final output. Compared to RF, XGBoost creates decision trees sequentially, with each new tree aiming to correct the errors made by the previous ensemble. It uses a gradient-based optimization approach to guide the tree-building process. Grid search will be used for hyperparameter tuning.

Internal validation

Five-fold repeated cross-validation will be used to account for optimism due to model overfitting [56]. After splitting development cohort data into five equally big parts, four parts are used as training set, and one part retained for testing model performance once model training is finished. This will be iterated until each of the parts will have been used for validation once. Random seeds will be set to make results reproducible.

Evaluation and comparison of model performance

The predictive performance of the developed models will be assessed and compared using measures of overall performance, calibration, and discrimination. To quantify overall classification performance, the distance between the predicted and actual outcomes will be evaluated using accuracy calculated in the confusion matrix. We define accuracy as the proportion of the predictions that are correct. The calibration ability of the developed model for the agreement between the estimated probabilities of the outcome and the observed outcome's proportion will be assessed via a calibration plot with loess smoother. To explore the model's ability to discriminate between those with and those without the outcome, the concordance (*c*) statistic will be used. For prediction models with binary outcomes, *c* is identical to the area under the receiver operating characteristic curve (ROC). The area under the curve (AUC) can be understood as the probability that the model assigns a higher probability of the outcome to a patient with the outcome compared to a randomly selected patient without the outcome [57]. Variable importance will be used to indicate how much each variable contributes to the predictive performance of all models using the *varImp* function from the caret package in R.

All analyses will be performed using R version 4.3.0 (R Foundation for Statistical Computing) [58].

Clinical usefulness of prediction models

We will perform decision curve analysis to evaluate the clinical utility and better understand whether the clinical use of a prediction model would do more good than harm [59]. By examining model performance against standardized net benefit and risk thresholds, one can assess how deciding to perform an intervention for an individual patient at their specific risk level, as indicated by the prediction model, compares with other management options such as performing an intervention for all

patients or no patients [60]. Our decision curve analysis will consider two clinically relevant scenarios: (1) Should total joint replacement surgery be recommended to patients who are at the decision point for surgical intervention? (2) Should patients who undergo total joint replacement surgery be provided more intense rehabilitation (e.g., increased pre- and postoperative rehabilitation programs) as a preventive measure to mitigate the risk for poor outcomes? Our decision curves will present the net benefit of our prediction model versus two default decisions that should always be present in a decision curve: treating none (with a net benefit of zero) and treating all.

External validation

We will externally validate our models using geographical validation as described before [30]. The performance of the models in the external validation cohorts will be assessed using measures of overall performance, discrimination, and calibration.

Model presentation

The final regression model consisting of the development and validation cohort will be presented using intercept to present baseline risk, regression coefficients, and odds ratios with 95% confidence intervals.

Discussion

We have described a protocol for developing, validating, and comparing traditional LR with different machine learning approaches for predicting patient outcomes and risk for revision surgery after THA and TKA using a large population-based cohort derived from a Swiss national joint registry. Prediction models as decision support tools may support preoperative clinical decision-making, health resource allocation, and care planning. To our knowledge, no risk prediction model using Swiss national data has been published in this area, and other studies comparing the predictive performance of ML algorithms are poorly reported [19].

A key strength of our study using registry data is the availability of relevant and comprehensive routine clinical data across a wide population, which enhances the generalizability and robustness of the prediction models. Limitations include the use of retrospective data collection, which limits the type of information that can be considered and included in the prediction models. Since patient-reported outcomes are only collected since 2019 in Canton Zurich, the study population for prediction of poor patient-reported outcomes will be limited to this population.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s41512-025-00200-z.

Additional file 1: Appendix.

Authors' contributions

LH and CAH will have full access to the data and will take responsibility for data integrity and accuracy of data analysis. LH and CAH led the conception and design of the protocol, with inputs from MAP, CB, CS, LR, MF and NS. LH and CAH drafted the protocol. All authors were involved in reviewing and revising the protocol. All authors read, provided feedback, and approved the final protocol. CAH had final responsibility for the decision to submit for publication.

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Data availability

The data that will be used and analyzed in this study are available only through the Swiss National Joint Registry. Restrictions apply to the availability of these data, which were used under license for the current study, and so are not publicly available.

Declarations

Ethics approval and consent to participate

The local ethics committee of Canton Zurich was consulted to assess the need for ethics approval (BASEC-Nr: 2024–01188). According to the ethics committee, this project was not subject to ethics approval given that the research group exclusively processed anonymous data. This is a register-based study where all data are anonymised and thus there is no ability to directly consent patients/participants.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Author details

[1]Musculoskeletal Epidemiology Research Group, University of Zurich and Balgrist University Hospital, Forchstrasse 340, Zurich 8008, Switzerland. [2]Epidemiology, Biostatistics and Prevention Institute (EBPI), University of Zurich, Zurich, Switzerland. [3]University Spine Centre Zurich (UWZH), Balgrist University Hospital, University of Zurich, Zurich, Switzerland. [4]Department of Advanced Computing Sciences, Maastricht University, Maastricht, the Netherlands. [5]Mathematics Centre Maastricht, Maastricht University, Maastricht, the Netherlands. [6]Center of Experimental Rheumatology, Department of Rheumatology, University Hospital Zurich, University of Zurich, Zurich, Switzerland. [7]Institute of Social and Preventive Medicine (ISPM), University of Bern, Bern, Switzerland. [8]ICES, Toronto, ON, Canada.

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